

Offline On-Policy Distillation under Agentic Shift

frozen data · no environment in the loop · no live teacher



Failure Diagnosis: a coverage-of-the-distilled-signal gap, not an env-access gap

The advantage is read at a single action $a \sim \pi_{\text{ref}}$, so the recovery action on the error branch is never distilled.

result: success 0.63, below the SFT floor

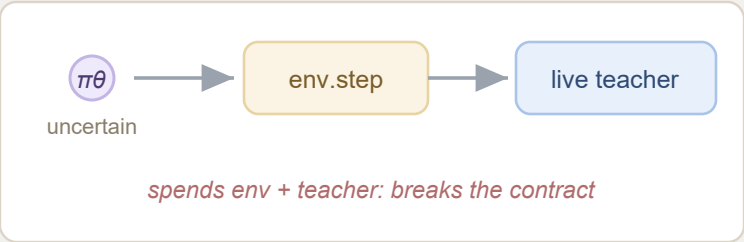
$\mathbb{E}[A \cdot \mathbb{I}[\log \pi]] \leftarrow \text{the support coverage assumption, as an IS condition}$

three remedies

Remedies

Patch 1 · Active Query (SafeDAgger)

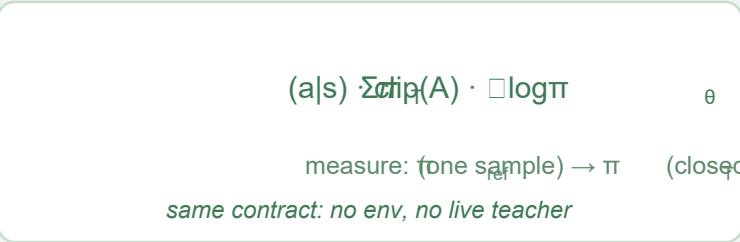
query teacher when $\max \pi_{\theta}(a|s) < \tau$



greedy success	0.88
env-steps	1558
teacher queries	159

Patch 2 · Full-distribution Advantage

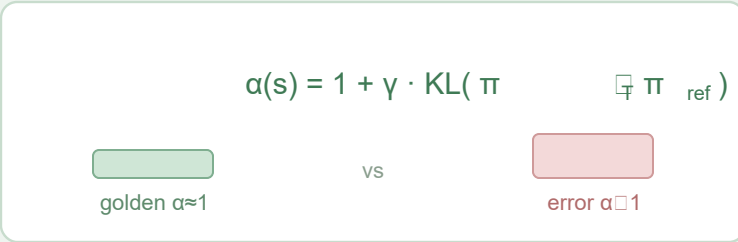
distill the whole teacher row, not one action



greedy success	1.00
env-steps	642
teacher queries	0

Patch 3 · Branch-aware Replay

reweight rare error-branch records



greedy success	1.00
sampled success	0.95
env-steps / queries	630 / 0

in-contract fixes: ceiling success at offline OPD's original budget